

DOCUMENT 06

Interface and user workflow

Make the actual cleaning footprint visible without removing the old controls

1. Mode controls and terminology

Proposed UI labels: “Current area (legacy)” and “Text-shaped + padding.” Final wording is open. The new mode must not be labeled as a guarantee of perfect detection or reconstruction.

Keep the current default initially. The user can choose text-shaped for a run or region. A run-level setting applies to newly prepared regions, not retroactively to every existing layer.

Core controls

Control	Behavior
Cleaning area	Switch between legacy and text-shaped.
Text-edge padding	Adjust the source-pixel margin around the base lettering mask.
Show affected area	Display the actual W footprint, not only bbox outlines.
Edit mask	Add/remove from the base or explicit correction layer.
Rebuild mask	Recompute from source evidence; preview before committing.
Engine	Existing cleanup engine selection; independent of area shape.

Detection profile and all-text/outside-bubble policy belong with run settings, not in the padding control. Recognition must not appear as a required step in the new workflow.

Visual contract

The filled/tinted overlay shows precisely which source pixels may be changed; an outline traces its edge. A bbox may still appear as a faint optional locator or selection handle, but it must be visually distinct from the affected-area overlay.

Keep mask opacity as a display preference only. It must not alter the cleaning strength or support. On low zoom, show a contour/locator for discoverability; at 100% allow an exact pixel view.

The current overlay uses bbox-positioned buttons and an accessible keyboard navigation model. Preserve those accessibility benefits while adding shape rendering and hit testing. [\[S16\]](#)

2. Selection, preview and correction

Region selection

A user can select a text group even if its footprint has several disconnected pieces. On pointer input, check the mask or a small display-only tolerance, not every pixel in a large bounding box. Keep keyboard and Layers-panel selection available for tiny masks.

A display-only hit tolerance must never expand the edit mask. If a selected region retains a rectangular keyboard focus ring, pair it with the true mask outline so the ring is not mistaken for the cleaning boundary.

Proposed interaction sequence

U1: choose cleaning-area mode and discovery/policy options.

U2: detect candidates and prepare text-shaped masks.

U3: inspect the actual affected-area overlay; show held/unsafe regions separately.

U4: adjust padding or correct the mask. Geometry updates immediately when derived from a cached base; a new segmentation pass shows a distinct preparing state.

U5: apply the selected engine through the displayed mask revision.

U6: compare with the original, correct residuals, or re-run without cumulative growth.

Corrections and tools

Support adding a missed outline or removing an artwork stroke. Manual mask edits are authoritative. Keep freehand/shape tools intact; merely selecting a text-shaped region must not reinterpret a user-drawn rectangle as automatic text.

Allow split/merge of selection groups without automatically filling the gaps between masks. Protect/exclude painting may be useful, but conflicts with known text must be visible rather than silently leaving letters behind.

Feedback and errors

Use specific messages: "Found; held by policy," "Text found; mask needs correction," "Optional SFX detector unavailable," "Preview changed; refresh before applying," or "Engine cannot use this source format." Do not collapse these into "No text found."

A mask with no trustworthy pixels cannot show a rectangle tinted as a ready-to-clean footprint. It remains a candidate marker with an explicit correction action.

3. Reference-specific acceptance and compatibility

Upper-right SFX

The preview should follow the Korean effect and its text-owned outline with the chosen margin. Avoid the large rectangular area over background figures and speed lines. Some background inside glyph counters or between strokes may remain untouched.

If the model cannot distinguish outline/glow from art, expose the uncertainty. This sample is a reason to test colored and outlined lettering, not proof that COO already handles it.

Lower bubble text

Keep the balloon outline outside the affected area. A compact padded footprint around the two text lines is acceptable; wiping the entire balloon or its surrounding rectangular crop is not the requested behavior.

A group footprint may connect nearby glyphs at the chosen radius. Do not connect the text to the balloon border through a thresholding artifact or broad automatic growth.

Old option

Switching back to legacy restores the old geometry workflow for subsequent operations. Existing layers are unchanged until explicitly rebuilt/reapplied. A chapter may contain legacy and text-shaped layers side by side.

The new mode should be usable with baseline models alone. The SFX detector is an independent optional enhancement. Neither a missing OCR file nor an uninstalled research segmenter should disable baseline text-shaped mask preparation when its required baseline models exist.

Proposed copy

Text-shaped + padding: "Clean within the detected text footprint and the margin you choose."

Padding: "Source-image pixels around the text. More padding includes more background."

Context: "Extra surrounding image used to estimate the background; it does not enlarge the affected area."

Uncertain mask: "Text was found, but its boundary needs correction before automatic cleaning."

These strings are draft copy. Implement through the app's translation catalog, preserve visible keyboard focus, and avoid changing the page position when a layer or toolbar control is used.

Done means: the user can distinguish detection bounds from the actual affected area, predict where an edit may occur, undo it, and continue using the old option unchanged.

Source IDs link directly to primary references. Full evidence ledger: Document 08.